Computer Science, B.S.

Location(s): On Campus

PROGRAM DESCRIPTION | PROGRAM STUDENT LEARNING OUTCOMES | ADMISSIONS, ENROLLMENT, & GRADUATION CRITERIA | PROGRAM COURSE REQUIREMENTS | HAVE QUESTION CONTACT US!

Program Description

The Bachelor of Science with a major in Computer Science program (BSCS) provides a blend of the foundations of computer science (CS) and applications in the information technology (IT) industry. The BSCS program emphasizes the study of computer systems architectu development, and data communications. Core technology areas include programming, computer architecture, operating systems, data communication, database systems, and software engineering. These areas are supported by a strong foundation in computing principles design of programming languages, data structures, and operating system principles. The program includes a mathematics component and mathematics concepts are incorporated into many of the major courses.

Graduates of the CS program are prepared for a variety of careers in CS and IT, especially in the development of software for distributed systems. Example job titles from KSU graduates of the CS program include information technology specialist, programmer analyst, sof engineer, network administrator, software developer, artificial intelligence engineer, back-end engineer/full stack developer, and software consultant. This program also prepares students for graduate studies in computing-related fields.

Program Student Learning Outcomes

Students who successfully complete this program will be able to:

1. Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

This program is a part of the College of Computing and Software Engineering.

This program offers a Double Owl Pathway! A Double Owl Pathway is an option for high-potential KSU undergraduate students who want to work towards a Master's degree while pursuing their Bachelor's degree at KSU. Click the owl for more information about pathway options and to learn if you qualify.

Accreditation

The Bachelor of Science with a major in Computer Science program is accredited by the Computing Accreditation Commission of ABET, https://www.abet.org/

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Admissions, Enrollment, and Graduation Criteria

Admissions Criteria

Admission to this program is open to all students who meet Kennesaw State University's general admission standards. Visit the Admissions section of the Catalog for more details.

Enrollment Criteria

This program does not have specific enrollment criteria; however, students are expected to meet the requirements of Academic Policy 4.0 ACADEMIC STANDING, DISMISSAL, & REINSTATEMENT.

Graduation Criteria

Each student is expected to meet the requirements outlined in Academic Policy 5.0 PROGRAM REQUIREMENTS & GRADUATION.

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Program Course Requirements

Core IMPACTS Curriculum (42 Credit Hours)

KSU's General Education Core IMPACTS Curriculum

Core IMPACTS Curriculum Specific to This Major

- Students should take MATH 1113 or higher in Mathematics & Quantitative Skills and MATH 1179 or higher in Applied Math.

- Students should take two four-hour laboratory sciences in core IMPACTS. Students may choose from CHEM 1211/1211L , CHEM 1212/1212L , PHYS 1111/1111L*, PHYS 1112/1112L , PHYS 2211/2211L*, PHYS 2212/2212L , BIOL 1107/1107L , or BIOL 1108/11

*Students cannot take both PHYS 1111/1111L and PHYS 2211/2211L nor PHYS 1112/1112L and PHYS 2212/2212L.

Core Field of Study (18 Credit Hours)

CSE 1321: Programming and Problem Solving I

CSE 1321: Programming and Problem Solving I

3 Credit Hours

Concurrent: CSE 1321L
Description
This course provides an introduction to computing with a focus on programming. Instruction centers on an overview of programming, problem-solving, and algorithm development. Particular topics include object-oriented design/programming, primitive data arithmetic and logical operators, selection and repetition structures, interactive user input, exception handling, using and designing basic classes, single-dimensional data structures with searching and sorting, and arrays. Programming assignments focus on tecl good programming style including proper documentation. The student is taught to efficiently design, code, and debug problem solutions and the relationship between correct code and security.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Apply computational methods to analyze data and solve real-world problems.
2. Demonstrate the appropriate use of data types and logical, relational, and arithmetic expressions in programs.
3. Construct algorithmic solutions that incorporate selection and repetition.
4. Implement collections appropriately to store and analyze data.
5. Define functions and classes in program solutions.

CSE 1321L: Programming and Problem Solving I Laboratory

CSE 1321L: Programming and Problem Solving I Laboratory

1 Credit Hours

Concurrent: CSE 1321
Description
Lab activities with programming to accompany CSE 1321.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Apply computational methods to analyze data and solve real-world problems.
2. Demonstrate knowledge of the syntax of a specific programming language.
3. Employ IDEs and compilers to learn how to compile, debug, run and test programs.
4. Demonstrate the appropriate use of data types and logical, relational, and arithmetic expressions in programs.
5. Construct algorithmic solutions that incorporate selection and repetition.
6. Implement collections appropriately to store and analyze data.
7. Define functions and classes in program solutions.

- CSE 1322: Programming and Problem Solving II

CSE 1322: Programming and Problem Solving II

3 Credit Hours

Prerequisite: CSE 1321 and CSE 1321L
Concurrent: CSE 1322L and (MATH 1113 or MATH 1190 or MATH 1179 or MATH 2202)
Description
The second course in computing provides coverage of more advanced topics of object-oriented programming. This includes the use of static variables and classes, non-linear data structures, inheritance and polymorphism, file input/output, exception handling, and parameterized types. Elementary data structures (linked lists, stacks, and queues) are introduced to solve application problems. Graphical user interfaces, parallel programming, database programming, and event-driven programming are also introduced. S will use good programming style including proper documentation.

- CSE 1322L: Programming and Problem Solving II Laboratory

CSE 1322L: Programming and Problem Solving II Laboratory

1 Credit Hours

Prerequisite: CSE 1321 and CSE 1321L
Concurrent: CSE 1322
Description
This course is the required and supervised lab course to accompany CSE 1322.

- MATH 2202: Calculus II

MATH 2202: Calculus II

4 Credit Hours

Prerequisite: MATH 1190 or (MATH 1179 and MATH 1189)
Description
This course is the second in the calculus curriculum and consists of two parts. The first part is concerned with the techniques of integration and applications of the integral. The second part is concerned with infinite sequences and series.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Evaluate a wide variety of standard integrals.
2. Apply integrals to a variety of problems involving area, volume, curve length, work done by a force and other similar quantities.
3. Understand the concept of convergence of improper integrals and be able to evaluate some basic improper integrals;
4. Understand the concept of convergence of infinite series and be able to derive and use a variety of tests for convergence of a series;
5. Understand the concept of a power series, be familiar with the important properties of power series, and understand the concept of the Taylor series of common functions.

- MATH 2345: Discrete Mathematics

MATH 2345: Discrete Mathematics

3 Credit Hours

Prerequisite: MATH 1113 or MATH 1190 or MATH 1179
Description
An introduction to the fundamentals of discrete mathematics. Topics include sets, formal logic, methods of proof, counting relations, functions, graphs and trees, and finite state automata.

Notes: Not intended for mathematics or mathematics education majors.

- TCOM 2010: Technical Writing

TCOM 2010: Technical Writing

3 Credit Hours

Prerequisite: ENGL 1102
Description
The course is an introduction to organization, style, and mechanics of technical writing. It includes practice in writing such typical documents as technical descriptions, instructions, proposals, and recommendation reports. Emphasis is placed on incorporating rl theory into planning, organizing, and writing reports; designing visual aids; and editing. Among other assignments, at least one complete technical report is required.

Major Requirements (40 Credit Hours)

Students must earn a "C" or better in these courses.

- CS 3305: Data Structures

CS 3305: Data Structures

3 Credit Hours

Prerequisite: (MATH 2345 or CSE 2300) and ((CSE 1322 and CSE 1322L), or MTRE 2710 with a "B" or higher, or CPE 3000 with a "B" or higher)
Description
This course introduces data structures, specification, application, and implementation. The case studies will illustrate how data structures are used in computing applications. The emphasis of the course is on linear and some nonlinear data structures and object principles. Topics include: abstract data types, stacks, queues, lists, binary search trees, priority queues, recursion, algorithm efficiency, trees, heaps, hash tables, and analysis of search and sort algorithms and their performance for implementation and manipulat programming language to be used in this course is any standard high-level object-oriented programming language such as C++, Java, and Ada.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Demonstrate understanding of single dimensional, multidimensional arrays, and dynamic arrays to store and access data, by applying these to programs.
2. Demonstrate learning of specifications and presentation of commonly used data structures by completely developing software. This must include appropriate documentation.
3. Demonstrate understanding and learn of advanced search and sort algorithms and their performance issues, by applying these to programs.
4. Demonstrate the analysis of the time complexity and space complexity of algorithms by explaining these concepts and applying these in programming.
5. Demonstrate the data structure covered in the courses by performing problem solving and application development.

- CS 3503: Computer Organization and Architecture

CS 3503: Computer Organization and Architecture

3 Credit Hours

Prerequisite: CSE 1322 and CSE 1322L

Description

The course covers computer architecture. Topics include data representation and encoding, binary arithmetic, fixed- and floating-point representation, numbering systems, error-control, instruction set architecture, assembly language and programming, microco memory organization and addressing, I/O, interrupts, internal architecture, instruction pipelining, multi-core architectures and CPU/ALU basics. The course also covers computer organization. Topics include Boolean Algebra, logic gates, flip-flops, counters, re combinational and sequential circuits, K-maps, circuit design, and various digital components.

- CS 3502: Operating Systems

CS 3502: Operating Systems

3 Credit Hours

Prerequisite: CS 3305 and CS 3503

Description

The course covers the basic concepts, design and implementation of operating systems. Topics include an overview of basic computing hardware components, operating system structures, process management, memory management, file systems, input/output protection and security. The Windows and/or UNIX/Linux operating systems will be reviewed as example systems.

- SWE 3313: Introduction to Software Engineering

SWE 3313: Introduction to Software Engineering

3 Credit Hours

Prerequisite: (CSE 1322 and CSE 1322L)

Description

This course provides an overview of the software engineering discipline, introducing the student to the fundamental principles and processes of software engineering. This course highlights the need for an engineering approach (both personal and team) to softv understanding of the activities performed at each stage in the development cycle. In this course, students will perform requirements analysis, design, implementation and testing. The course presents software development processes at the various degrees of gra Students will become aware of libraries of standards (IEEE, ACM, SWEBOK, etc.).

- CS 3410: Introduction to Database Systems

CS 3410: Introduction to Database Systems

3 Credit Hours

Prerequisite: CSE 1322 and CSE 1322L

Description

Introduction to database management systems, database processing, data modeling, database design, development, and implementation. Particular emphasis is placed on the relational approach to database management and processing, which focuses more on t nature of a database than its physical characteristics. Relational database programming assignments are drawn from the fields of business. Includes implementation of current DBMS tools and SQL. Ethical and security topics related to databases will be introduc

- CS 4306: Algorithm Analysis

CS 4306: Algorithm Analysis

3 Credit Hours

Prerequisite: CS 3305

Description

Algorithm analysis using formal and mathematical techniques and fundamental algorithm design strategies are studied. Topics include asymptotic analyses of complexity bounds using big-O, little-o, omega, and theta notations. The algorithmic strategies (brute greedy, divide-and-conquer, recursive backtracking, dynamic programming, branch-and-bound, heuristics, and reduction) are covered. Also included are standard graph and tree algorithms. Additional topics include standard complexity classes, time-and-spac in algorithms, and analyzing both recursive algorithms and non-recursive (iterative) algorithms.

- CS 3622: Fundamentals of Data Communications

CS 3622: Fundamentals of Data Communications

3 Credit Hours

Prerequisite: A grade of "B" or better in both CSE 1322 and CSE 1322L.
Description
An introduction on the fundamentals and underlying theory of data communication networks, their architecture, principles of operations and performance analyses. Topics include OSI reference model and standards, communication codes, network protocol cor synchronous and asynchronous transmission methods, line coding, signaling, effects of bandwidth and noise, digital and analog modulation, interfacing, error/flow/media-access control, switching and routing methods, and network topologies. Other areas stu Local area networks (LANs), Wide area networks (WANs), Wireless networks, Fiber optic networks, internetworking technologies, and an introduction to the Internet, TCP/IP, cryptography and network security.

- CS 4504: Parallel and Distributed Computing

CS 4504: Parallel and Distributed Computing

3 Credit Hours

Prerequisite: CS 3305 and CS 3503
Concurrent: CS 3502
Description
This course introduces students to the fundamental principles common to the design and implementation of programs that run on two or more interconnected computer systems - in parallel or distributed configurations. Topics to be covered include: essentials c operating systems, network protocols for process communication, and synchronization using message queues; understanding of client-server paradigms, web-based group or collaborative communication systems; advanced distributed computing paradigms fo computing and handling concurrency issues; and sockets. Programming will focus on using API's for parallel or distributed applications (e.g., MPI and RMI).

- CS 4308: Concepts of Programming Languages

CS 4308: Concepts of Programming Languages

3 Credit Hours

Prerequisite: CS 3305 and CS 3503
Description
This course covers the fundamental concepts on which programming languages are based and the execution models supporting them. Topics include values, variables, bindings, type systems, control structures, exceptions, concurrency, and modularity. Languag representing different paradigms are introduced.

- CSE 3801: Professional Practices and Ethics

CSE 3801: Professional Practices and Ethics

2 Credit Hours

Prerequisite: (CSE 1322 and CSE 1322L) or IT 3123 (may take concurrently)
Description
This course covers the historical, social and economic consideration of the discipline. It includes studies of professional conduct, risks, and liabilities, and intellectual property relative to the software engineering and computing professions. Software engineering/computing case studies will be used.

- CS 4850: Computer Science Senior Project

CS 4850: Computer Science Senior Project

3 Credit Hours

Prerequisite: CS 3502 and SWE 3313

Description

This course provides a capstone experience for CS majors to promote a successful transition into the work place or further academic study. Students will have the opportunity to practice essential project management skills and work with current software tools a technologies. Student teams will develop a project scope, project plan, document functional specifications, develop a design document, implement specified functions, provide weekly progress reports, give project presentations to the class, conduct final project presentation to the instructor and/or project sponsor, and provide a complete final report that includes documentation of all class activities. Each team will designate a team leader who is responsible for coordinating work tasks, team meetings, communications instructor and/or project sponsor, and team effort.

- STAT 2332: Probability and Data Analysis

STAT 2332: Probability and Data Analysis

3 Credit Hours

Prerequisite: MATH 1190 or MATH 1179

Description

This course is a foundational, calculus-based introduction to statistics and probability. The following conceptual themes will be developed through the process of statistical investigation: exploratory data analysis (univariate and bivariate), fundamentals of exper design and sampling, planning and conducting a study, exploring random phenomenon using probability and simulation, and the fundamentals of statistical inference. Technology is integrated into each theme, and the statistical software package used will be cl the instructor.

- MATH 3260: Linear Algebra I

MATH 3260: Linear Algebra I

3 Credit Hours

Prerequisite: MATH 1190 or (MATH 1179 and MATH 1189)

Description

An introduction to linear algebra and some of its classical and modern applications. Among topics to be included will be systems of linear equations, matrices, determinants of matrices and applications, vector spaces, and inner product spaces. Significant use of technology will be employed in performing matrix computations.

One (1) credit hour carried over from Applied Mathematics.

One (1) credit hour carried over from Technology, Mathematics, and Sciences.

Major Electives or Concentration (15 Credit Hours)

Complete a series of major elective courses OR select one major concentration and complete its requirements.

Major Electives (15 Credit Hours)

Complete a minimum of 9 credit hours of CS 3000-4000 level coursework and 0-6 credit hours from the following list of courses:

- SWE 3633: Software Architecture and Design

SWE 3633: Software Architecture and Design

3 Credit Hours

Prerequisite: SWE 3313 or CPE 3000

Description

This course covers the fundamental design principles and strategy for software architecture and design. Architectural styles, quality attributes, design notations and documents, reference architecture, domain specific architecture in architecture process and p oriented design, component-oriented design, and interface design in detailed design process are discussed.

- SWE 3643: Software Testing & Quality Assurance

SWE 3643: Software Testing & Quality Assurance

3 Credit Hours

Prerequisite: SWE 3313 or CPE 3000
Description
This course will show how software quality assurance and configuration management are performed and how software process improvement is maintained in order to assure the highest possible quality. Topics include software process metrics and their use testing approaches, methods and techniques. Development of QA plans, reviews, inspections and audits will be done.

- SWE 3683: Embedded Systems Analysis and Design

SWE 3683: Embedded Systems Analysis and Design

3 Credit Hours

Prerequisite: CS 3305
Description
The analysis and design course focuses on using modern methods, techniques, and tools for specification and design of embedded systems. Topics include analytical methods such as RMA, development methods such as HOOD, and notations like UML, Pet are covered. Performance evaluation based on modeling and simulation techniques is also covered. This is a project based course.

- SWE 4633: Cloud Software Development

SWE 4633: Cloud Software Development

3 Credit Hours

Prerequisite: CS 3305
Description
This course discusses the fundamental concepts and tools of cloud-based application development. Students will learn how to develop apps on top of the cloud with an understanding of various contemporary cloud providers (e.g., Amazon Web Services, Go Cloud, or Microsoft Azure). Students will do hands-on projects focused on building cloud-run applications using their platform of choice.

- CSE 4983: CSE Computing Internship

CSE 4983: CSE Computing Internship

3 Credit Hours

Prerequisite: Senior standing
Description
This course helps students gain practical experience through real-world projects and professional work. Students demonstrate an ability to apply computing principles and technologies relevant to their major in a specific real-world project jointly supervised industry mentor and a faculty advisor. Students work in a project team in an enterprise environment demonstrating ethical behavior as a computing professional, an understanding of social, professional and ethical issues related to computing, and an ability integrate the knowledge acquired in preceding courses. Communication skills and leadership are also evaluated as well as professional computing skills and knowledge. 150+ hours per semester required at an internship site. The course can not be repeated f

Artificial Intelligence Concentration

Required Courses (12 Credit Hours)

AI 3642: Artificial Intelligence

AI 3642: Artificial Intelligence

3 Credit Hours

Prerequisite: CS 3305 or AI 2000

Description

The primary objective of this course is to provide a introduction to the basic principles and applications of Artificial Intelligence. It covers the basic areas of artificial intelligence including problem solving, knowledge representation, reasoning, decision m planning, perception and action, and learning – and their applications. Students will design and implement key components of intelligent agents of modern complexity and evaluate their performance. Students are expected to develop familiarity with cu research problems, research methods, and the research literature in AI.

Course Learning Outcomes

Students who successfully complete this course will be able to:

1. Demonstrate knowledge of AI knowledge representation.
2. Demonstrate understanding of advanced search and reasoning algorithms.
3. Design an AI system with Lisp or Prolog for problem solving.

AI 4267: Machine Learning

AI 4267: Machine Learning

3 Credit Hours

Prerequisite: AI 3642

Description

This course provides a broad introduction to machine learning and statistical pattern recognition including supervised, unsupervised, and ensemble learning. Topics include K-NN, Naïve Bayes Classifier, parametric and non-parametric methods, suppor machines, kernel machines, neural networks, clustering, dimensionality reduction, and model evaluation. The learning theory including bias/variance tradeoffs and large margins will be introduced. This course will also discuss recent applications of ma learning such as data mining, autonomous navigation, speech recognition, and text and web data processing.

- AI 4732: Machine Vision

AI 4732: Machine Vision

3 Credit Hours

Prerequisite: AI 3642

Description

This course introduces concepts and techniques in machine vision. Students successfully completing this course will be able to apply a variety of image processing techniques for the design and analysis of efficient algorithms for real-world applications, optical character recognition, face detection and recognition, motion estimation, human tracking, and gesture recognition. Topics include basic image enhancement, corner and edge detection, image morphology, linear and non-linear filters, image transfo camera models, two-dimensional and three-dimensional image geometry, clustering and segmentation (and classification), object recognition and Bag-of-Words models, image texture, shape analysis, and tracking.

- AI 4742: Natural Language Processing

AI 4742: Natural Language Processing

3 Credit Hours

Prerequisite: AI 3642

Description

This course provides an introduction to the field of natural language processing (NLP). The topics include creating systems that can understand and produce language for applications such as information extraction, machine translation, automatic summ question-answering, and interactive dialogue systems. This course covers linguistic (knowledge-based) and statistical approaches to language processing in the areas of syntax (language structures), semantics (language meaning), and pragmatics/discou interpretation of language in context). Students will design and develop programs for analyzing and extracting information from large online corpora.

Elective Courses (3 Credit Hours)

Select 3 credit hours from the following list of courses:

- AI 4277: Deep Learning

AI 4277: Deep Learning

3 Credit Hours

Prerequisite: AI 3642
Concurrent: AI 4267
Description
This course covers the foundations of Deep Learning; how to build neural networks and how to design successful deep learning projects. The course topics include convolutional networks, sequence modeling such as recurrent and recursive neural netwo (RNNs), long short-term memory (LSTM), Adam, Dropout, BatchNorm, Xavier/He initialization, state-of-the-art technologies, and research topics leveraging Deep Learning. The course includes programming assignments in Python and in TensorFlow.

- CS 4491: Advanced Topics in Computer Science (in concentration)

CS 4491: Advanced Topics in Computer Science

3 Credit Hours

Prerequisite: A grade of "C" or better in any prerequisite course. Prerequisite course(s) vary depending upon the topic.
Description
This course provides the current and relevant topics in an advanced Computer Science area of interest to faculty.

Notes: It may substitute for a CS major elective.

- CS 4492: Undergraduate Research

CS 4492: Undergraduate Research

3 Credit Hours

Prerequisite: Department Permission and (CS 4265 or AI 4267 or CS 4270 or CS 4322 or CS 4412 or CS 4504 or CS 4512 or CS 4514 or CS 4522 or CS 4523 or CS 4524 or CS 4612 or CS 4622 or CS 4632 or CS 4712 or CS 4720 or CS 4722 or AI 4732).
Description
This course promotes undergraduate research in Computer Science. Students develop research ideas and conduct research work to investigate topics aligned with the department's research plan and with the guidance of Computer Science faculty membe Students document their findings in final reports, present their findings, and prepare research papers for publication in appropriate venues.

CSE 4983: CSE Computing Internship (in concentration)

CSE 4983: CSE Computing Internship

3 Credit Hours

Prerequisite: Senior standing
Description
This course helps students gain practical experience through real-world projects and professional work. Students demonstrate an ability to apply computing principles and technologies relevant to their major in a specific real-world project jointly supervi industry mentor and a faculty advisor. Students work in a project team in an enterprise environment demonstrating ethical behavior as a computing professional, an understanding of social, professional and ethical issues related to computing, and an ab integrate the knowledge acquired in preceding courses. Communication skills and leadership are also evaluated as well as professional computing skills and knowledge. 150+ hours per semester required at an internship site. The course can not be repeat credit.

Data Science Concentration

Required Courses (12 Credit Hours)

CS 4265: Big Data Analytics

CS 4265: Big Data Analytics

3 Credit Hours

Prerequisite: CS 3305 and (CS 3410 or CSE 3153)
Description
This course covers algorithms and tools that are needed to build MapReduce applications with Hadoop or Spark for processing gigabyte, terabyte, or petabyte-sized datasets on clusters of commodity hardware. A wide range of data algorithms will be di
this course.

- CS 4412: Data Mining

CS 4412: Data Mining

3 Credit Hours

Prerequisite: CS 3305 and (CS 3410 or CSE 3153)
Description
This course covers fundamental data mining concepts and techniques for discovering interesting patterns from data in various applications. Topics include data preprocessing, data warehousing and OLAP, mining frequent patterns, classification, clusteri
tend analysis.

- CS 4422: Information Retrieval

CS 4422: Information Retrieval

3 Credit Hours

Prerequisite: CS 3305 and (CS 3410 or CSE 3153)
Description
This course covers the fundamentals of Internet search engines, including Web crawlers, inverted indices, hyperlink analysis, and relevance ranking. Also covered are advanced topics including information extraction for knowledge base construction, qu
answering, search marketing and ad targeting, and activity mining for relevance optimization and personalization.

- CS 4522: HPC & Parallel Programming

CS 4522: HPC & Parallel Programming

3 Credit Hours

Prerequisite: CS 4504
Description
This course will introduce parallel programming techniques for shared memory and distributed memory systems. Topics include threading, OpenMP, and MPI.

Elective Courses (3 Credit Hours)

Select 3 credit hours from the following list of courses:

- CS 4524: Cloud Computing

CS 4524: Cloud Computing

3 Credit Hours

Prerequisite: CS 4504
Description
This course discusses the fundamental concepts and techniques of cloud computing. Students will develop an understanding of cloud computing architecture, Infrastructure as a Service (IaaS), Platform-as-a-Service (PaaS), Software as a Service (SaaS), Virtualization, and Application Development on Cloud.

- CS 4722: Computer Graphics and Multimedia

CS 4722: Computer Graphics and Multimedia

3 Credit Hours

Prerequisite: CS 3305
Description
The basic principles and practices of interactive computer graphics and multimedia systems are covered in this introductory course. The design and implementation of state-of-the-art computer graphic rendering and visual multimedia systems are the m the course. The sub-topics of the course deal with specific input/output hardware devices and their technology, software and hardware standards, programming methods for implementing 3-dimensional graphical applications and interactive multimedi applications, and a study and evaluation of the effectiveness of graphic/multimedia communications. A large component of the class is the building of a large-scale application.

- CS 4491: Advanced Topics in Computer Science (in concentration)

CS 4491: Advanced Topics in Computer Science

3 Credit Hours

Prerequisite: A grade of "C" or better in any prerequisite course. Prerequisite course(s) vary depending upon the topic.
Description
This course provides the current and relevant topics in an advanced Computer Science area of interest to faculty.

Notes: It may substitute for a CS major elective.

- CS 4492: Undergraduate Research

CS 4492: Undergraduate Research

3 Credit Hours

Prerequisite: Department Permission and (CS 4265 or AI 4267 or CS 4270 or CS 4322 or CS 4412 or CS 4504 or CS 4512 or CS 4514 or CS 4522 or CS 4523 or CS 4524 or CS 4612 or CS 4622 or CS 4632 or CS 4712 or CS 4720 or CS 4722 or AI 4732).
Description
This course promotes undergraduate research in Computer Science. Students develop research ideas and conduct research work to investigate topics aligned with the department's research plan and with the guidance of Computer Science faculty membe Students document their findings in final reports, present their findings, and prepare research papers for publication in appropriate venues.

- CSE 4983: CSE Computing Internship (in concentration)

CSE 4983: CSE Computing Internship

3 Credit Hours

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Cyber and Network Security Concentration

Required Courses (12 Credit Hours)

- CS 3626: Cryptography

CS 3626: Cryptography

3 Credit Hours

Prerequisite: MATH 2345 or CSE 2300
Concurrent: CS 3305
Description
The course covers both mathematical and practical foundations of cryptography. Topics include basic number theory for cryptography, conversion of text, and implementation using a programming language. The course includes historical cryptography, cryptography, asymmetric cryptography, hash functions, and well-known attack strategies with countermeasures. Exercises cover programming of simple cryptography in a programming language.

- CS 4612: Software Security

CS 4612: Software Security

3 Credit Hours

Prerequisite: CS 3502 and CS 3626
Description
The course introduces the fundamental concepts and principles of software security. Topics covered include buffer overflows, defense mechanisms, return oriented programming, reverse engineering, vulnerabilities analysis. Additional topics: mobile sec hardware platform security, embedded system security.

- CS 4622: Computer Networks

CS 4622: Computer Networks

3 Credit Hours

Prerequisite: CS 3503 and CS 3622
Description
This course covers computer networking and includes software application-related, protocol-related and security-related issues involved in the Internet. Topics include basic network structures, mechanisms for application-to-application communications layering, Internet addressing, unicast and multicast routing, connection establishment and termination, data flow and congestion control, and error handling. A specific protocol suite will be examined in detail. More advanced topics that build on the stu understanding of network protocols are also introduced, such as network security, mobile networks and the future Internet.

- CS 4626: Computer and Network Security

CS 4626: Computer and Network Security

3 Credit Hours

Prerequisite: CS 3626 and CS 4622

Description

This course is a comprehensive study of the security principles and practices for computer systems and networks. Topics to be covered include basic security concepts, common network attack techniques and impacts, common security policies, Internet p vulnerabilities, basic cryptographic tools and secure protocols. Defense techniques such as authentication, access control, encryption, and network intrusion detection will be discussed. This course will also survey new and emerging network security top applications and technologies. Exercises for this course will include network programming in some language and using various tools in understanding and analyzing packet traces and network traffic.

Elective Courses (3 Credit Hours)

Select 3 credit hours from the following list of courses:

CS 4491: Advanced Topics in Computer Science (in concentration)

CS 4491: Advanced Topics in Computer Science

3 Credit Hours

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Description

This course promotes undergraduate research in Computer Science. Students develop research ideas and conduct research work to investigate topics aligned with the department's research plan and with the guidance of Computer Science faculty membe Students document their findings in final reports, present their findings, and prepare research papers for publication in appropriate venues.

CSE 4983: CSE Computing Internship (in concentration)

CSE 4983: CSE Computing Internship

3 Credit Hours

Prerequisite: Senior standing

Description

This course helps students gain practical experience through real-world projects and professional work. Students demonstrate an ability to apply computing principles and technologies relevant to their major in a specific real-world project jointly supervi industry mentor and a faculty advisor. Students work in a project team in an enterprise environment demonstrating ethical behavior as a computing professional, an understanding of social, professional and ethical issues related to computing, and an ab integrate the knowledge acquired in preceding courses. Communication skills and leadership are also evaluated as well as professional computing skills and knowledge. 150+ hours per semester required at an internship site. The course can not be repeat credit.

IT 4823: Information Security Administration & Privacy

IT 4823: Information Security Administration & Privacy

3 Credit Hours

Prerequisite: (CSE 2300 or MATH 2345) and (IT 3123 or CS 3503)
Description
The student develops knowledge of the principles of information assurance at the policy, procedural, and technical levels to prepare the student for a role as a business decision-maker. Real-world examples from the text and current events will be used to demonstrate the applicability of the techniques of information assurance.

- IT 4833: Wireless Security

IT 4833: Wireless Security

3 Credit Hours

Prerequisite: IT 4323 or ECET 3400 or CS 4622
Description
This course covers methods and techniques to secure wireless networks against threats and attacks. The topics may include Security and Cryptography, Network Security Protocols, Security and Layered Architecture, Voice-Oriented Wireless Networks, D Oriented Wireless Networks, Security in Traditional Wireless Networks, Security in Wireless LAN, Security in Wireless Ad Hoc Network, Special Topics such as Mobile Security.

- IT 4843: Ethical Hacking for Effective Defense

IT 4843: Ethical Hacking for Effective Defense

3 Credit Hours

Prerequisite: IT 4323 or ECET 3400 or CS 4622
Description
This course focuses on detection of network and system vulnerabilities by taking an attacker-like approach to system, network, and data access. Topics include network attacks and defenses, Operating System and application vulnerabilities, social engine attacks, and malware. Ethical, legal implications of network attacks are also discussed.

- IT 4853: Computer Forensics

IT 4853: Computer Forensics

3 Credit Hours

Prerequisite: IT 4323 or CS 4622 or ECET 3400
Description
This course studies techniques and tools in computing investigation, digital evidence collection, recovery, and analysis. Topics include: Legal issues relating to digital evidence, recover deleted files and discover hidden information, reconstruct user activi mail, temporary Internet files and cached data, assess the integrity of system memory and process architecture to reveal malicious code.

- IT 4883: Infrastructure Defense

IT 4883: Infrastructure Defense

3 Credit Hours

Prerequisite: IT 4323 or ECET 3400 or CS 4622
Description

This course provides an overview of the infrastructure assessment and penetration testing process and the processes and techniques for improving the defensibility of that infrastructure.

University Electives (5 Credit Hours)

In accordance with KSU Graduation Policy, students must earn a grade of "D" or better in these courses while maintaining a minimum 2.00 cumulative GPA.

Free Electives (5 Credit Hours)

Select 5 credit hours of 1000-4000 level coursework from the University Catalog.

Program Total (120 Credit Hours)